# CodeGhar

**Your Code. Every Version. Always Recoverable.**

GIRASE MAYUR
202403103510603

## 1. Project Overview

CodeGhar is a lightweight, web-based version control system for text and source code files. It allows users to upload, create, edit, and track multiple versions of files with an intuitive interface. The system provides a GitHub-like edit view, version restation, and secure file management without complex Git commands.

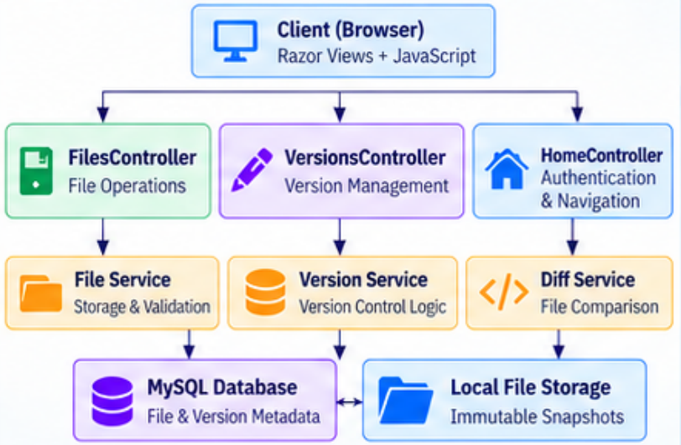
## 2. Key Features

- **Upload & Create** — Add text and code files with strict validation
- **Restore Versions** — Revert to any previous version without losing history
- **Edit & Save** — Automatic version creation on every change
- **Code Diff Viewer** — Line-by-line comparison with highlighted changes
- **Version History** — Track all versions with timestamps and change summary

## 3. Technology Stack

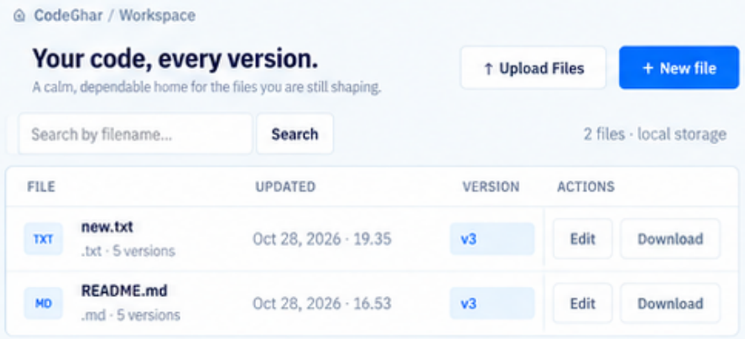
- ASP.NET Core MVC (C#)
- Razor Views
- MySQL Database
- Entity Framework Core
- HTML5, CSS3, JavaScript
- Visual Studio Code (Monaco Editor)
- Myers Diff Algorithm
- Bootstrap (UI Components)

## 4. System Architecture

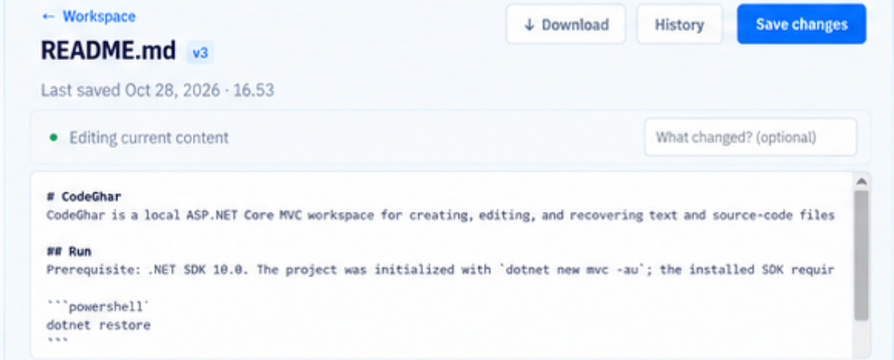



## 5. Application Screenshots

### 1 File Dashboard




### 2 Code Editor




### 3 Version History




## 6. Supported File Types

| | |
|---|---|
| **Text Files** | .txt, .md, .markdown |
| **Web Files** | .html, .css, .js, .ts, .jsx, .tsx |
| **Source Code** | .cs, .java, .py, .c, .cpp, .php, .go |
| **Config Files** | .json, .xml, .yaml, .yml, .gitignore |
| **Others** | .sql, .sql, .bat, .ps1, .rs, .vue |

## 7. Version Lifecycle

v1 File Created → v2 Edited & Saved → v3 More Changes → v4 Restore Previous Version

Restoring a version creates a new version without deleting existing history.

## 8. Security & Validation

- Only text and code files allowed
- File size limit (2 MB)
- Strict file extension validation
- Path traversal protection
- Files stored outside wwwroot
- User authentication and authorization
- Antiforgery protection for all actions

## 9. Use Cases

- Manage and track code snippets
- Maintain notes and documentation
- Keep versioned configuration files
- Collaborate in academic projects
- Learn version control concepts easily

## 10. Project Impact

- Simplifies version control for students
- Easy-to-use, intuitive interface
- No need for complex Git commands
- Helps in learning and managing code history
- Useful for academic and personal projects
- Lightweight and runs locally